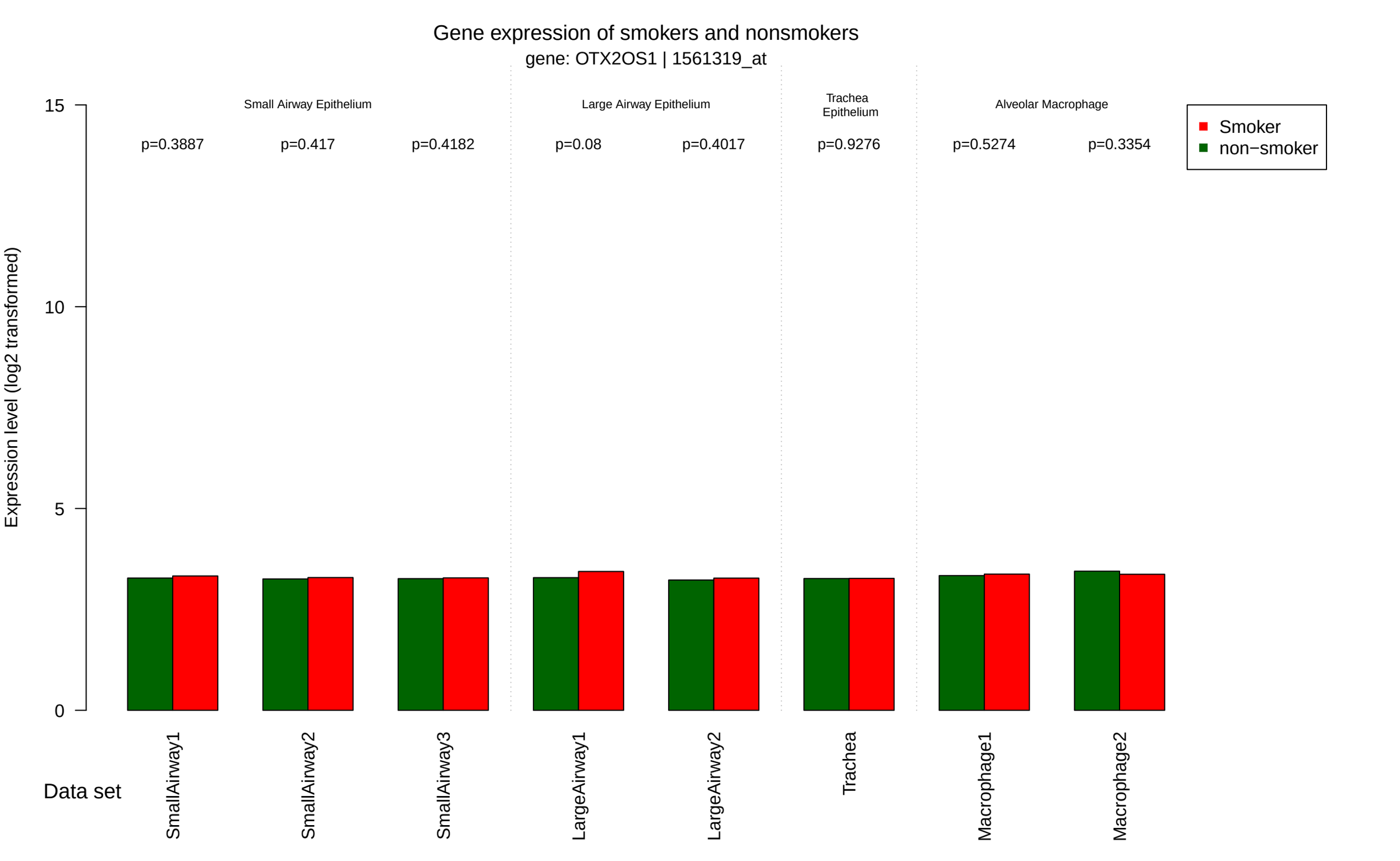
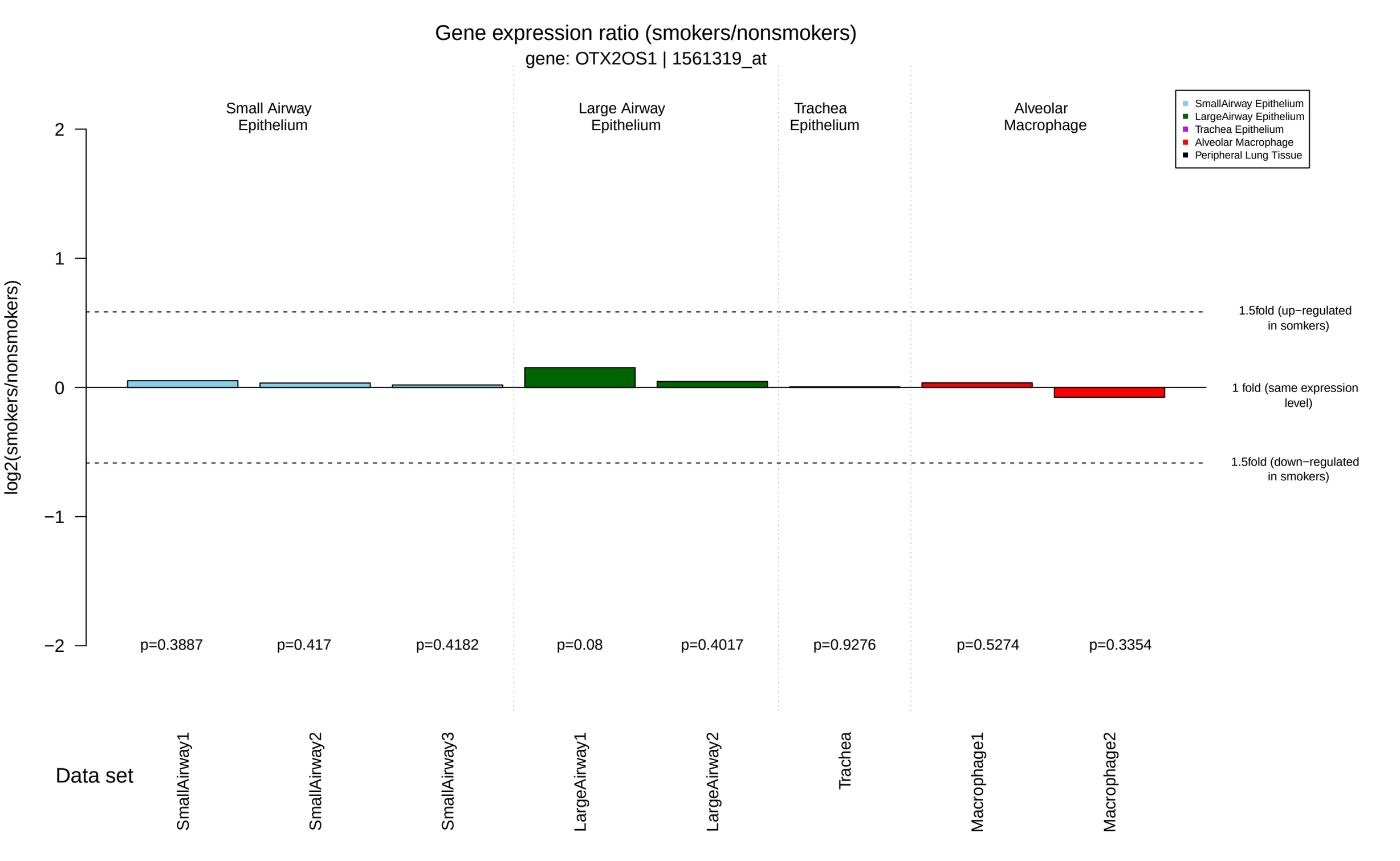
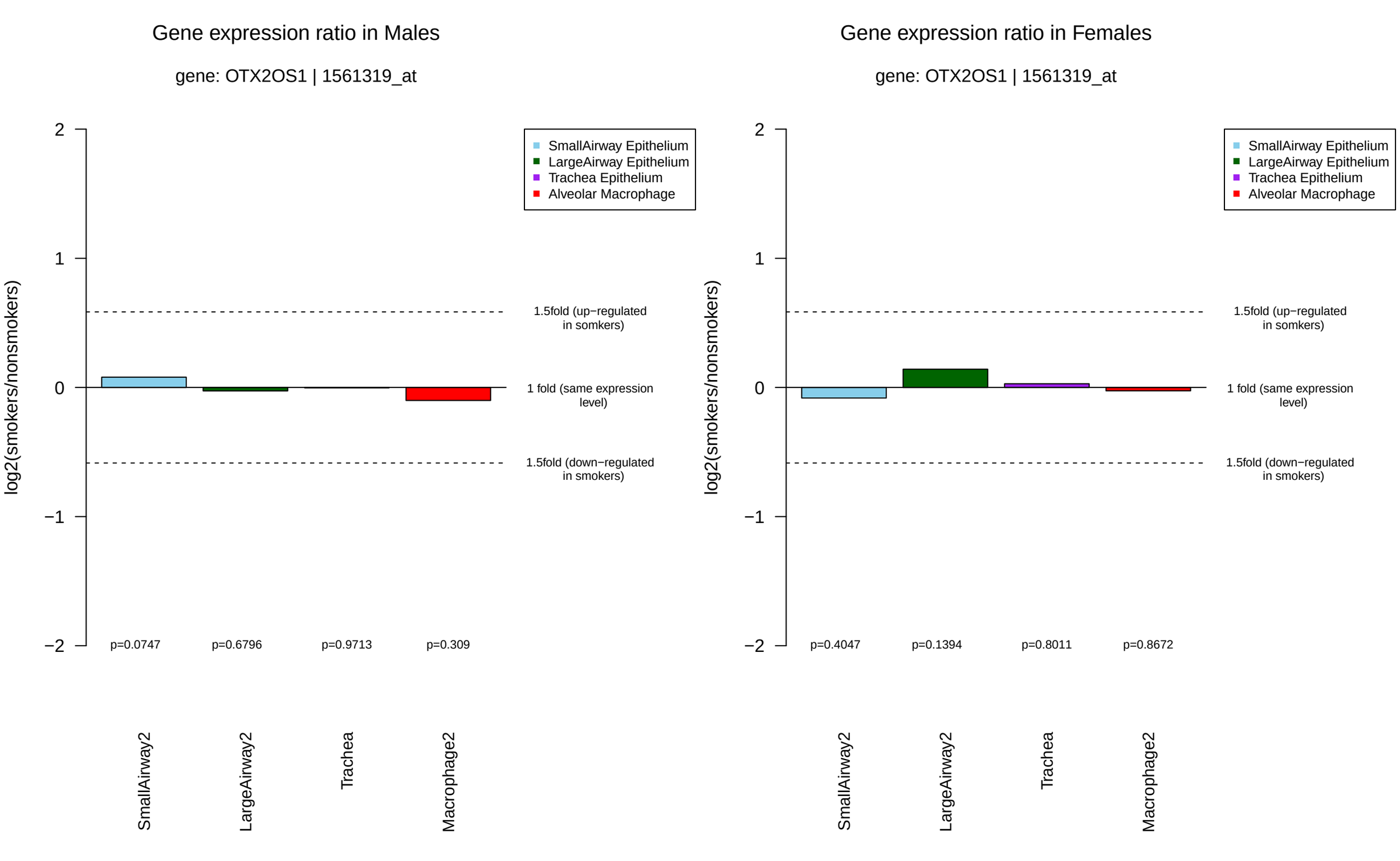
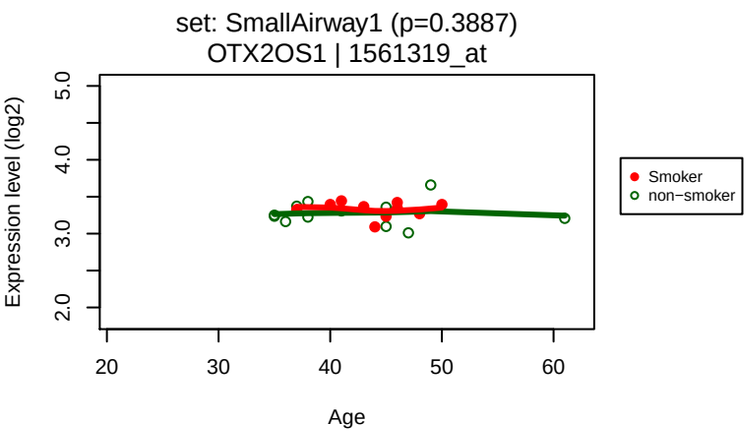
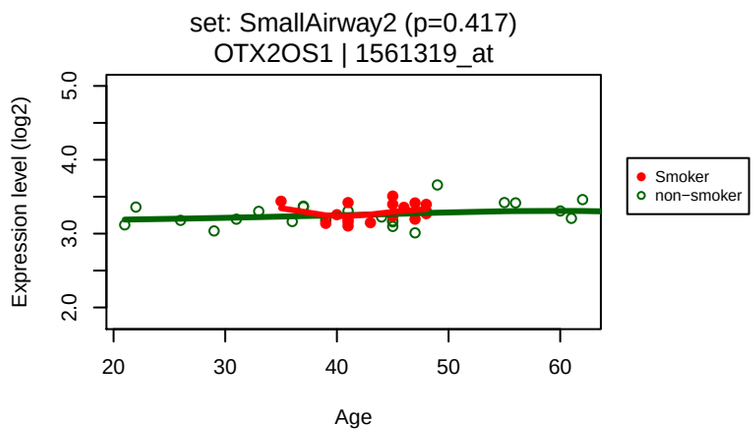
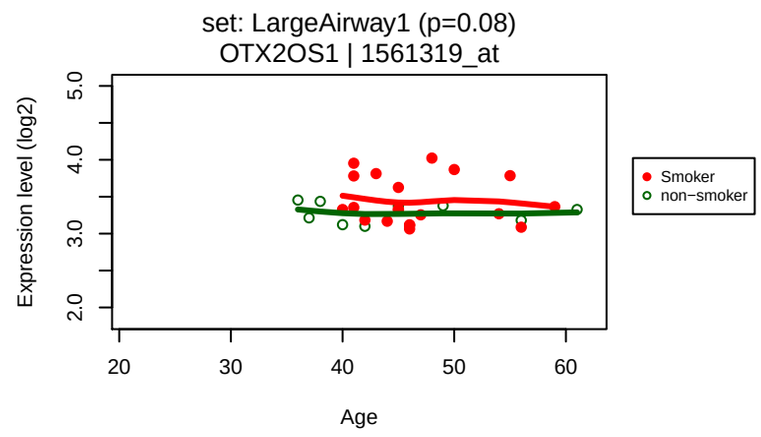

Gene expression of smokers and nonsmokers
gene: OTX2OS1 | 1561319_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3887
p=0.417
p=0.4182
p=0.08
p=0.4017
p=0.9276
p=0.5274
p=0.3354
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

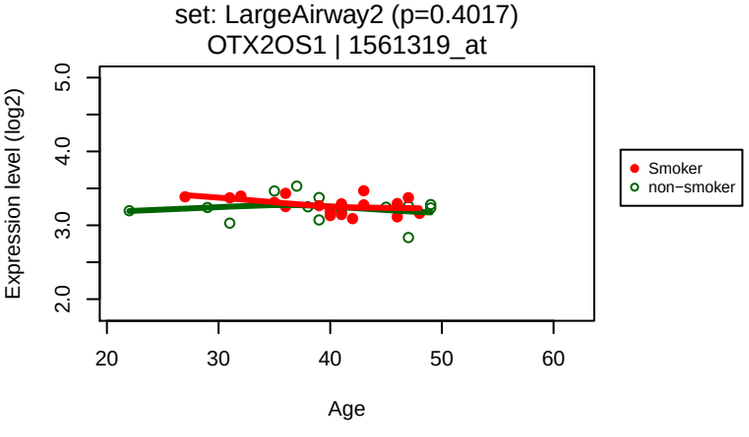

Gene expression ratio (smokers/nonsmokers)
gene: OTX2OS1 | 1561319_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3887
p=0.417
p=0.4182
p=0.08
p=0.4017
p=0.9276
p=0.5274
p=0.3354
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

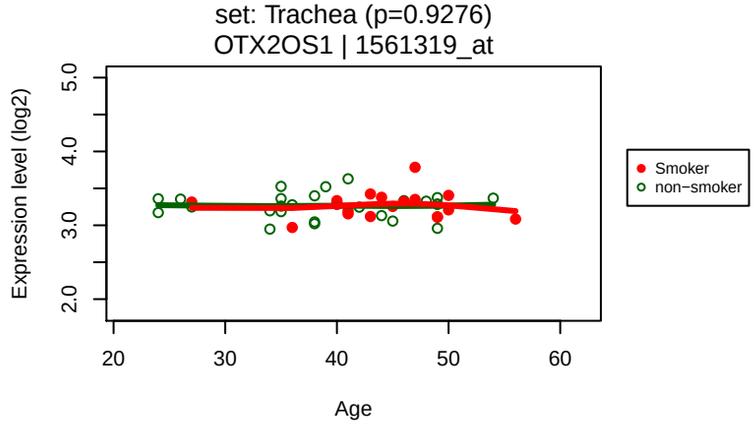

Gene expression ratio in Males
gene: OTX2OS1 | 1561319_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0747
p=0.6796
p=0.9713
p=0.309
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OTX2OS1 | 1561319_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4047
p=0.1394
p=0.8011
p=0.8672
SmallAirway2
LargeAirway2
Trachea
Macrophage2

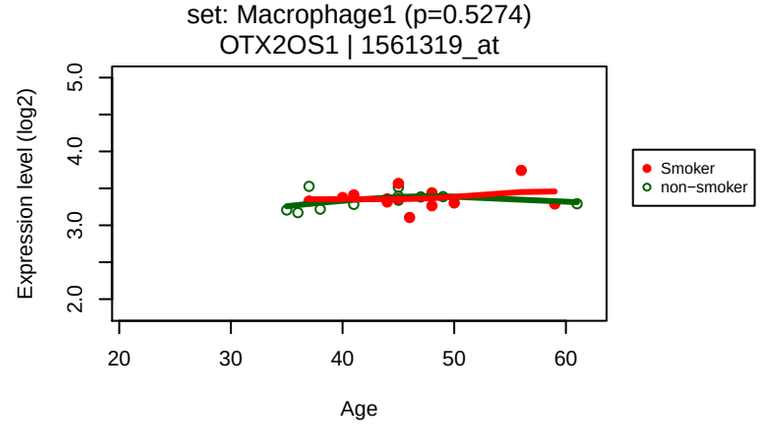

set: SmallAirway1 (p=0.3887)
OTX2OS1 | 1561319_at
Expression level (log2)
Age
Smoker
non−smoker

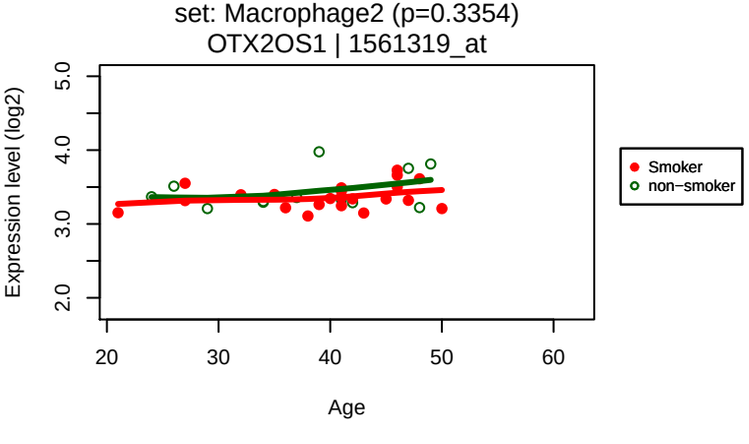

set: SmallAirway2 (p=0.417)
OTX2OS1 | 1561319_at
Expression level (log2)
Age
Smoker
non−smoker

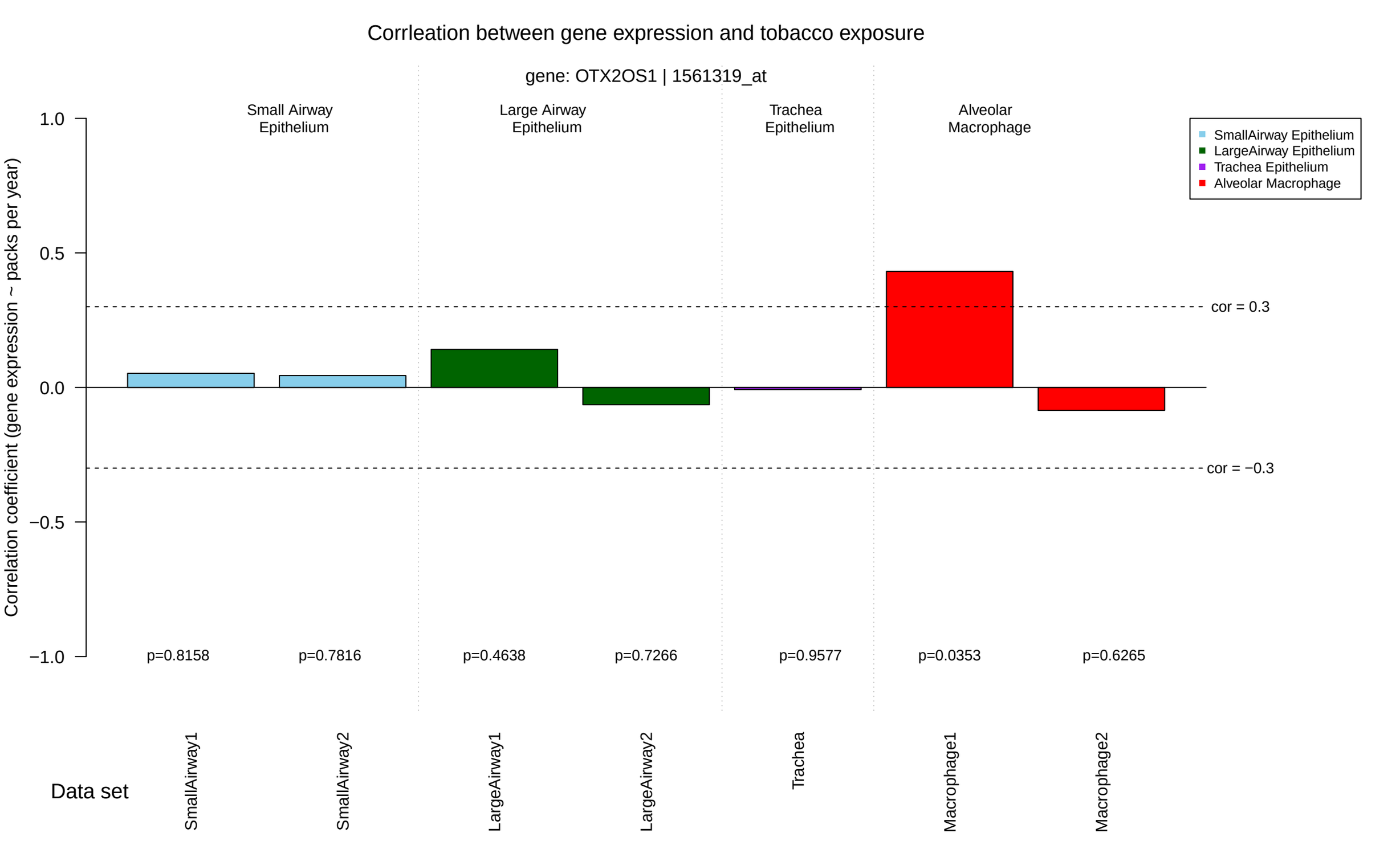

set: LargeAirway1 (p=0.08)
OTX2OS1 | 1561319_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4017)
OTX2OS1 | 1561319_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9276)
OTX2OS1 | 1561319_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5274)
OTX2OS1 | 1561319_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3354)
OTX2OS1 | 1561319_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OTX2OS1 | 1561319_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.8158
p=0.7816
p=0.4638
p=0.7266
p=0.9577
p=0.0353
p=0.6265
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2